

Version Control & CI/CD Troubleshooting Guide

For AI Engineering Students (RAG & Ollama Projects)

1. Common Git Issues on Windows + Git Bash

Problem	Cause	Solution
'Only one usage of each socket address (each protocol family)	Ollama already running in background	Stop Ollama from system tray or kill process
ConnectionError: Failed to connect to Ollama (localhost:11434) vs IPV4	See Ollama (localhost:11434) vs IPV4	Use Ollama CLI Bash (host='http://127.0.0.1:11434') or
invalid path ... colon (:) in folder name	Windows doesn't allow ':' in filenames (e.g. intro:CRIPS4)	git config core.protectNTFS false + sparse-checko
fatal: destination path '.' already exists	Trying to clone into non-empty folder	clone to a new empty folder first
You are in a sparse checkout...	Using git sparse-checkout	git sparse-checkout disable (or fix with protectNTFS
warning: adding embedded git repository to	Nested .git folder (e.g. rag-intro inside another dep)	git add --force or copy files to c
remote origin already exists	Trying to add remote again	git remote set-url origin <your-url>

2. Sparse Checkout – How to Pull Only One Folder

Best commands we used:

1. git clone --no-checkout <url> .
2. git sparse-checkout init --cone
3. git sparse-checkout set "Folder Name"
4. git config core.protectNTFS false ← Important for Windows + colons
5. git checkout

3. Connecting & Pushing to Your Own Repository

- git remote -v → check current remotes
- git remote set-url origin https://github.com/yourusername/yourrepo.git
- git branch -M main
- git push -u origin main

4. Best Practices for AI / RAG Projects

- Never commit virtual environments (Alenv, venv)
- Use .gitignore (add __pycache__/, *.pyc, .env, venv/)
- Keep large models & data outside Git (use .gitignore or Git LFS)
- Run ollama serve in a separate terminal / use Ollama desktop app
- Test embeddings with explicit Client(host='http://127.0.0.1:11434')
- Create clean folders instead of nesting repos

Generated from real troubleshooting session • April 2026